Earth and Planetary Science Letters, 79 (1986) 441–452
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

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The palaeomagnetism of the main zone in the western Bushveld Complex

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Received November 20, 1985; revised version accepted June 19, 1986

Sampling was done at thirty-four sites in the main zone of the western Bushveld Complex, South Africa. Alternating field (AF) and thermal demagnetization of specimens indicated a multicomponent NRM in the majority of specimens. A primary magnetization direction ($D = 357.8°$; $I = 70.9°$; $\alpha_{95} = 2.8°$; $k = 80$) was recovered from thirty-two sampling sites. Fold tests indicate that the igneous layering was emplaced in a horizontal position and that the NRM predates tilling of the layered sequence. Two sampling sites situated in the lower part of the main zone yielded magnetization directions approximately antipodal to the above mentioned magnetization direction. The palaeomagnetic pole position (9.2°N; 27.3°E), calculated for the main zone of the western Bushveld Complex, indicates that this part of the main zone had acquired its magnetization later than the main zone in the eastern Bushveld Complex and later than the underlying critical zone.

1. Introduction

Recent palaeomagnetic investigations on certain layered intrusions such as Sudbury [1,2] and Skaergaard [3], have shown that palaeomagnetic data can be an important indicator of the tectono-intrusive history of layered intrusions. For this reason, a comprehensive palaeomagnetic investigation of the Bushveld Complex was initiated in order to extend the study of Gough and Van Niekerk [4] and to relate the new palaeomagnetic results from the layered sequence to more modern concepts of the genesis and structure of the Complex [5–7].

In this report palaeomagnetic data from the main zone of the western Bushveld Complex are presented. The geographical and stratigraphical positions of thirty-four sampling sites are shown in Figs. 1 and 2. The majority of sites is situated in quarries and road cuttings. The resultant relatively unweathered samples enabled the extraction of consistent and stable magnetization directions from all sampling sites and it was unnecessary to reject any data.

2. Laboratory measurements

Samples were taken with a petrol driven portable core drill. The core samples are 25 mm in diameter and up to 250 mm in length. Four core samples were taken at each sampling site. Each core was cut into five or more 25 mm long cylindrical specimens. Core orientation was done with a sun compass [8] which allows orientation to a precision of 1° [9].

The remanent magnetization of specimens was measured using a Digico "Complete Results" fluxgate spinner magnetometer. In practice it was found that to measure a reliable direction of magnetization to within one degree, a minimum intensity of magnetization of 0.5×10^{-3} A m^{-1} was required.

Continuous thermal demagnetization of specimens was done with a slightly-modified form of the spinner magnetometer. It was possible to measure remanence at high temperatures without the need for cooling the specimen in a field free space. Operating the instrument in this mode, however, led to a decreased sensitivity, and it was found that the minimum intensity of magnetization that could be measured reliably was in the region of 4×10^{-3} A m^{-1}.

Alternating field (AF) demagnetization of specimens was done with an apparatus built at the Geological Survey of South Africa following a description by McElhinny [10]. This apparatus is capable of creating a peak alternating magnetic field of 120 mT.

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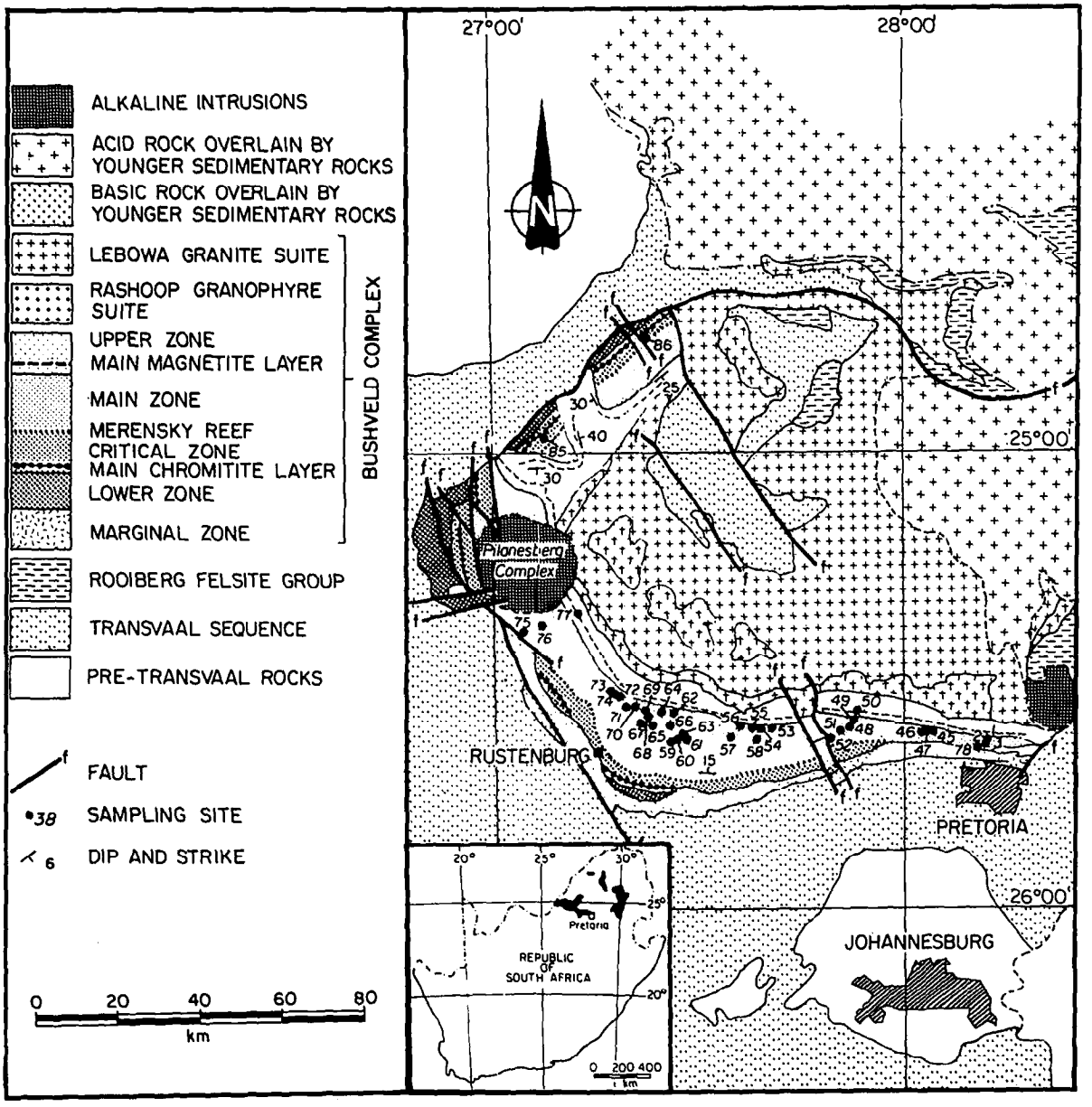


Fig. 1. Geological map of the western Bushveld Complex showing sample site positions.

3. Natural remanent magnetization

With the exception of specimens from sites 75 and 76, specimens from all the sites have a fairly high intensity of NRM. The average intensity of the NRM is 2078×10^{-3} A m^{-1}, which is slightly lower than the average intensity of the NRM of the main zone in the eastern Bushveld Complex [11].

The NRM directions of thirty two sites, together with the site statistics, are listed in Table 1, while the magnetization directions are shown in Fig. 3A. Magnetization directions of specimens from sites 48 and 51 were random at 95% confidence limits, using the test by Watson [12]; accordingly, these two sites were rejected from the analysis of NRM data.

There are thirty sites with NRM directions

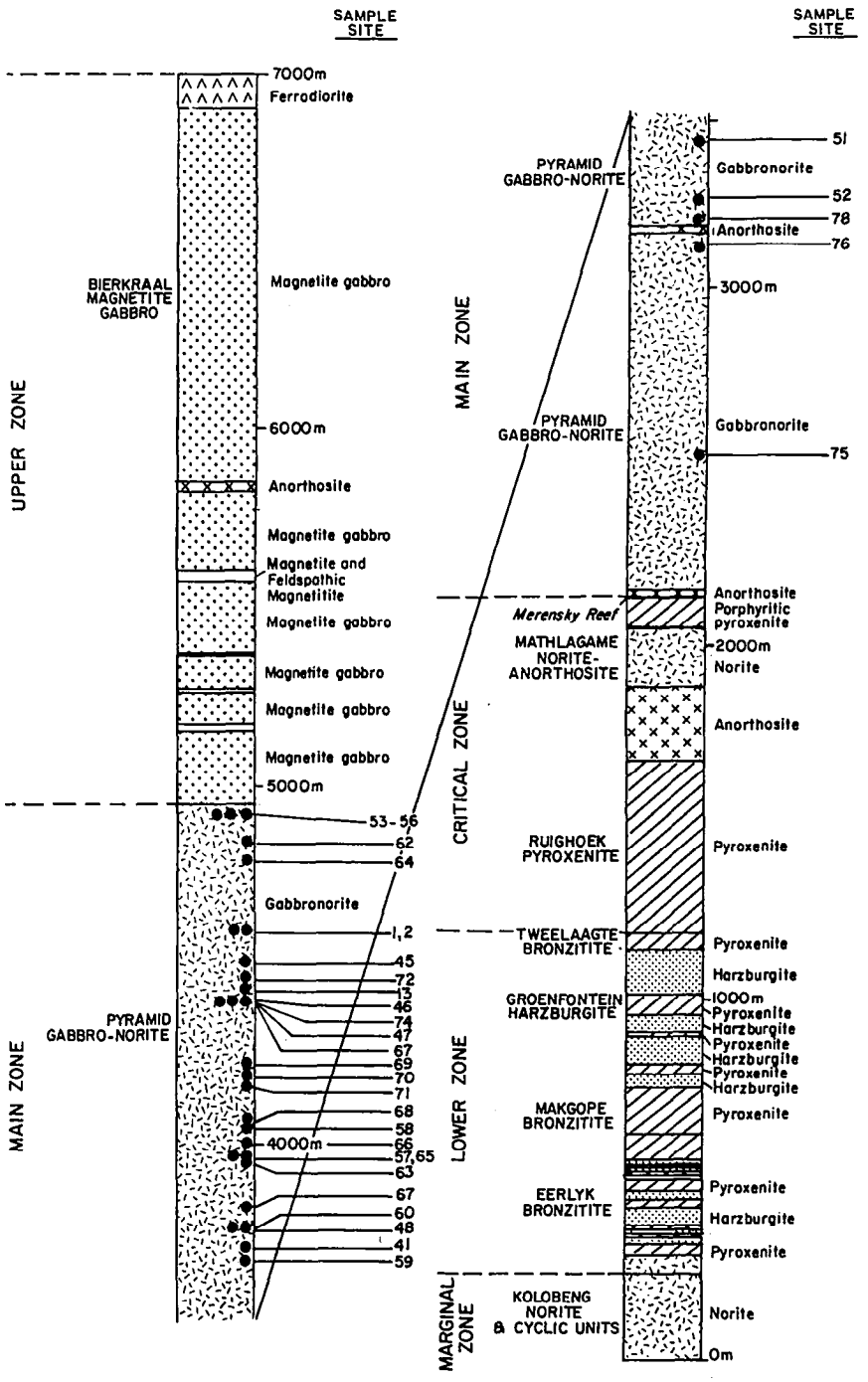


Fig. 2. Lithostratigraphic column of the mafic layered sequence in the western Bushveld Complex. Stratigraphic positions of the sampling sites are indicated.

reversed (positive inclination). These sites form group A: $N = 30$; $D = 345.8°$; $I = 79.5°$; $\alpha_{95} = 3.7°$; $k = 52$.

Group A exhibits a smeared distribution in a mainly north-northwesterly direction, but a number of magnetization directions lies along a north-northeast trend (Fig. 3A). This may be due to the dip of the igneous layering at each site, or due to a

TABLE 1

Natural remanent magnetization directions of sites in the main zone (A: western Bushveld Complex; B: with the igneous layering in a horizontal position)

Site	N	A				B			
		D (°)	I (°)	α_{95} (°)	k	dip (°)	dip direction (°)	D (°)	I (°)
1	4	215.3	64.8	9.6	167	20	N10	264.0	79.0
2	4	319.9	81.8	2.6	1259	20	N10	355.5	64.0
45	4	270.6	87.6	3.7	612	25	N10	4.3	65.3
46	4	274.2	87.2	5.7	180	25	N10	4.3	65.1
47	4	143.5	86.9	3.0	893	25	N10	15.7	67.0
49	4	313.6	89.0	9.7	161	14	N10	6.7	75.0
52	4	12.9	66.8	10.9	72	15	N355	6.4	53.0
53	4	326.6	85.0	5.1	582	10	N350	342.2	75.3
54	4	342.8	82.3	3.6	663	10	N350	346.8	72.3
55	4	321.0	85.6	3.9	554	10	N350	341.1	76.0
56	4	43.4	77.0	3.0	956	25	N355	12.0	55.1
57	4	348.5	82.5	4.7	381	25	N05	1.0	57.8
58	4	0.5	72.3	5.5	280	10	N03	2.0	62.3
59	4	1.4	69.6	4.0	526	8	N20	6.3	61.9
60	4	46.6	66.6	6.8	183	5	N20	42.0	62.0
61	4	346.9	67.7	6.3	216	5	N20	352.2	62.4
62	4	346.1	67.2	3.8	582	6.5	N22	353.4	61.7
63	4	358.9	73.8	7.7	142	10	N25	8.5	64.5
64	4	299.8	82.4	4.6	402	15	N20	354.9	72.0
65	4	366.2	72.0	6.4	204	5	N25	8.5	64.5
66	4	342.2	68.3	0.6	9839	5	N25	345.8	68.4
67	4	332.3	68.0	6.0	233	7	N30	344.5	63.6
68	4	326.5	75.4	3.4	743	8	N30	347.4	70.5
69	4	356.7	73.5	3.0	953	8	N30	7.1	66.4
70	4	325.7	80.5	2.7	1201	7.5	N35	355.4	75.9
71	4	314.8	79.0	2.0	2094	10	N40	354.4	74.5
72	4	336.4	75.7	4.8	366	15	N40	8.2	65.2
73	4	329.5	77.7	4.5	343	10	N40	0.1	71.7
74	4	345.7	69.9	3.3	772	10	N40	2.3	62.9
75	4	153.4	−54.0	5.9	257	20	N70	180.0	−51.6
76	4	182.3	−49.8	5.7	257	20	N70	199.0	−39.5
78	4	75.4	83.8	13.9	79	20	N05	20.2	67.2

N is the number of core samples taken at each site; D is the declination of the magnetization direction, measured eastwards from north; I is the inclination of the magnetization direction; downwards positive, upwards negative; α_{95} is the semi-angle of cone of 95% confidence; k is the estimate of the precision parameter κ.

multi-component NRM.

The magnetization directions of two sites, situated in the lower part of the main zone (Fig. 2), have negative inclinations (sites 75 and 76, Table 1 and Fig. 3A) and lie in a roughly antipodal position to group A.

The NRM directions of the sites, with the igneous layering rotated to a horizontal position, as well as the dip and dip direction of the layering at each site, are listed in Table 1 and shown in Fig. 3B.

After correction for dip, group A now has the following statistics: $N = 30$; $D = 0.5°$; $I = 67.6°$; $\alpha_{95} = 3.0°$; $k = 78$.

The ratio of the estimates of the two precision parameters k_1/k_2, is 1.5 and this improvement in k after application of the fold test [13], is not significant at the 95% confidence limit according to the criterion of McElhinny [14]. However, if the improvement is tested at the 90% confidence limit, the improvement in k is significant. Unfortunately the newer fold test devised by McFadden and Jones [15] cannot be applied to data from only one limb of the Bushveld Complex. Because

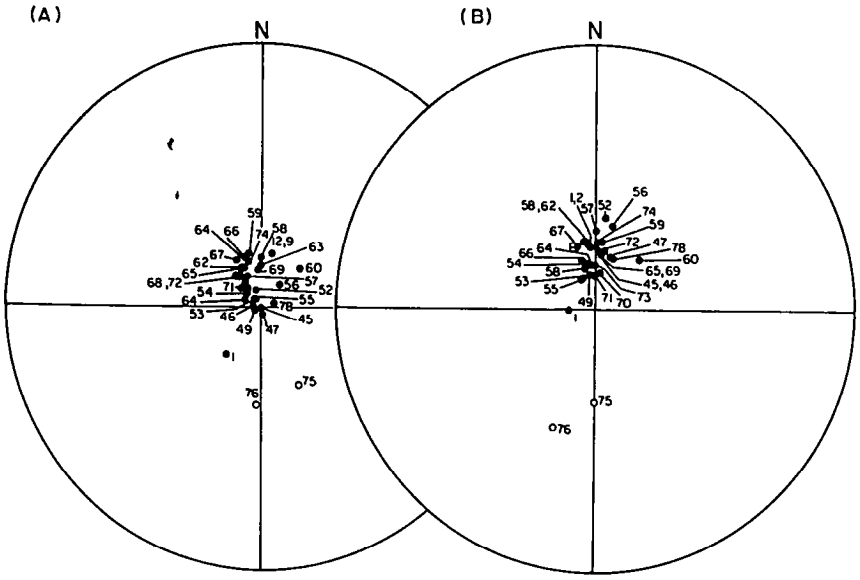


Fig. 3. A. Equal angle projections of NRM directions of sites. B. NRM directors from sites in the main zone, western Bushveld, with the igneous layering in a horizontal position. ● = lower hemisphere, ○ = upper hemisphere.

the dip values used in this test are only accurate to within a few degrees, it is reasonable to accept the result of the fold test as an improvement, which implies that the igneous layering acquired its present orientation after the main zone in the western Bushveld had cooled to a temperature below its Curie point.

4. Bulk alternating field demagnetization

Bulk AF demagnetization of specimens of each site was done at an optimum AF value indicated by the Briden SI [16] for each site. The results of the bulk AF demagnetization are presented in Table 2 and the magnetization directions of the

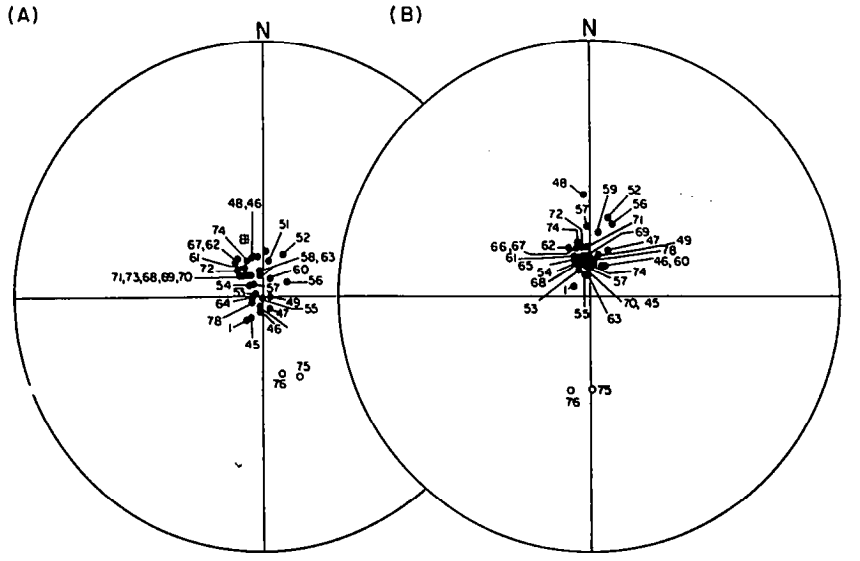


Fig. 4. Magnetization directions from sites in the main zone, western Bushveld Complex, (A) after bulk AF demagnetization and (B) with the igneous layering in a horizontal position.

TABLE 2

Magnetization directions of sites in the main zone, western Bushveld Complex (A: after bulk AF demagnetization; B: with the igneous layering in a horizontal position)

Site	A					B	
	Demagnetization field (mT)	D (°)	I (°)	α_{95} (°)	k	D (°)	I (°)
1	20	214.8	77.8	8.0	236	304.0	81.7
2	40	193.1	86.4	4.7	386	9.3	73.6
45	60	209.9	79.6	4.2	483	356.8	74.4
46	40	182.7	83.9	4.7	256	12.4	71.0
47	50	141.3	84.1	3.9	555	22.1	68.5
48	60	352.5	71.8	13.1	50	357.1	46.9
49	30	87.2	85.8	1.6	2970	25.5	74.5
51	30	11.1	73.5	20.6	20	9.9	75.5
52	30	27.0	68.8	5.6	268	14.6	55.2
53	10	303.0	86.6	4.9	358	338.5	77.4
54	20	320.0	82.7	3.9	532	337.3	73.3
55	20	185.5	89.6	3.7	605	349.2	80.4
56	30	61.0	76.6	3.6	640	18.0	57.3
57	20	334.0	84.0	2.9	973	358.9	59.7
58	40	353.6	78.7	4.7	388	358.8	68.8
59	20	5.0	69.5	3.4	731	9.0	61.7
60	60	25.2	80.4	2.6	1226	23.5	75.4
61	70	330.2	74.6	6.3	213	341.5	71.0
62	30	336.0	72.7	1.8	2650	347.7	67.7
63	50	354.6	80.6	6.6	197	354.6	80.6
64	20	270.0	85.9	2.2	1778	4.0	75.9
65	30	328.6	75.0	6.0	231	341.5	71.7
66	30	342.2	72.1	1.3	5304	350.8	68.2
67	30	323.2	70.6	2.4	1425	338.7	67.0
68	30	319.5	77.8	2.9	950	345.9	73.4
69	30	332.4	79.4	3.0	892	356.5	73.7
70	30	323.4	79.2	2.2	1676	351.4	75.0
71	30	312.3	76.4	2.4	1503	347.2	72.9
72	30	317.2	74.2	3.3	775	356.3	67.0
73	30	317.8	76.7	4.9	346	351.2	72.4
74	30	325.9	70.1	5.1	314	347.9	65.5
75	30	153.5	−52.1	14.0	43	178.5	−49.9
76	30	165.8	−56.3	4.5	414	191.4	−49.7
78	30	243.0	84.9	12.0	57	359.7	72.2

sites are shown in Fig. 4A.

After AF demagnetization, the magnetization directions of sites 48 and 51 are consistent at sample and site levels. Group B now forms, which includes directions from sites 48 and 51: $N = 32$, $D = 336.1°$; $I = 82.0°$; $\alpha_{95} = 3.4°$; $k = 58$.

Although two additional sites are included in this group, it is not a significant improvement on the NRM data. This is further reflected by the tendency of some sites to show weaker in-site grouping of magnetization directions after bulk AF demagnetization, which is apparent from smaller k values in Table 2.

With the igneous layering in a horizontal position, the statistics of group B change to: $N = 32$, $D = 357.8°$; $I = 70.9°$; $\alpha_{95} = 2.8°$; $k = 80$. Although the value of k has increased from 58 (before correction for dip) to 80 (after correction for dip), the improvement is not significant at confidence limits as low as 90%, using the criterion of McElhinny [14].

The previous results indicate that either the Briden SI [16] failed in the majority of cases to locate a primary magnetization direction, or that at least some of the stable NRM originated during or after tilling.

The above conclusion does not hold true for sites 75 and 76, the only two sites where the magnetization directions are normal. Figs. 3 and 4 show that these two directions grouped closer together after bulk AF demagnetization, but it is not certain whether the rotation of the igneous layering to the horizontal, has resulted in a significantly improved grouping of these two directions.

5. Stepwise alternating field demagnetization of pilot specimens

During stepwise AF demagnetization of specimens at progressively higher AF values, magnetization directions from virtually all the sites in group B exhibited a slow change in direction. At low AF values the magnetization vectors remain stable with very little movement, but at AF values higher than approximately 40 mT the directions start to change, with the directional change and divergence of vectors becoming larger as the AF values increase. This type of response is illustrated in Fig. 5 with the aid of specimens from sites 56 and 62. From this it must be concluded that the magnetization directions in groups A and B are not primary magnetization directions, but rather resultant magnetization directions due to primary and secondary magnetization.

In Fig. 6 remagnetization circles [17] fitted to AF demagnetization data from four sites, are shown. A comparison of the orientation of these remagnetization circles and the dip and dip directions of the layering at each site (Table 1) indicates that there is no correlation between the secondary magnetization(s) and the dip of the igneous layering at each site.

The magnetization component responsible for the grouping of the magnetization vectors into group B, dominates the magnetic characteristics of the specimens and this primary magnetization component almost certainly was acquired by all the specimens at more or less the same time. With this and the previous arguments on the effect of the dip of the igneous layering in mind, only one plausible explanation for the diverging circles of remagnetization can be postulated, namely that the secondary magnetization component was acquired at a different time at each site, resulting in a different direction of secondary magnetization for each site. It must, however, be emphasized

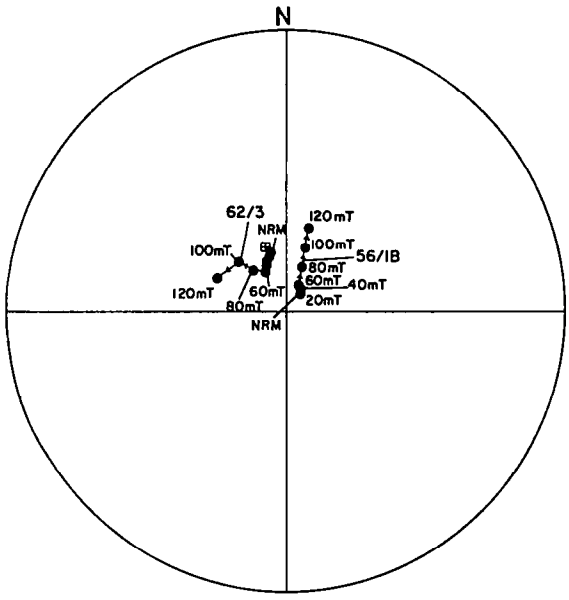


Fig. 5. The directional response of two specimens from sites 56 and 62 to stepwise AF demagnetization.

that this explanation is heavily dependent on the accuracy of the measured dip and dip direction values of the igneous layering.

The direction of the primary magnetization vector can possibly be recovered by the use of intersecting remagnetization circles, following the technique of Halls [17]. To do this, a number of assumptions are made:

(1) The primary magnetization component was acquired with the igneous layering in a horizontal position.

(2) The primary magnetization component has

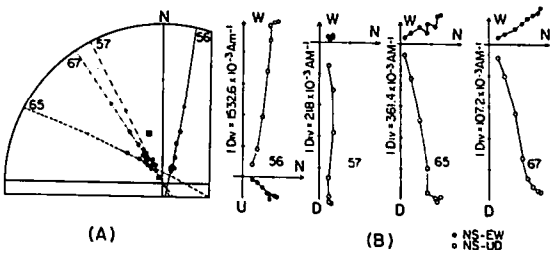


Fig. 6. A. The directional response of four specimens from sites 56, 57, 65 and 67 to stepwise AF demagnetization. Lines represent circles of remagnetization fitted to the respective data sets. B. Data presented as Zijderveld orthogonal vector plots.

a lower coercivity than the secondary magnetization components.

(3) Sets of remagnetization circles will always define more than one intersection point on a sphere, but in this particular case, the correct intersection point yielding the direction of the primary magnetization component, will be the point from which the magnetization vectors diverge during stepwise AF demagnetization.

Circles of remagnetization (great circles) were fitted with a least squares technique [18] to data corrected for dip, obtained from specimens during stepwise AF demagnetization. The remagnetization circles are shown in Fig. 7 and the calculated intersection point yields the direction of the primary magnetization vector with $D = 354.7°$ and $I = +72.0°$. This direction does not differ much from the mean direction of group B, but it is corrected for the effect of the secondary magnetization components.

Difference vectors, corrected for the dip of the igneous layering, representing the removed vector between each successive AF demagnetization step, are plotted in Fig. 8 for four sites. At very low AF values the vectors removed are different for each site; thereafter, at progressively higher AF values,

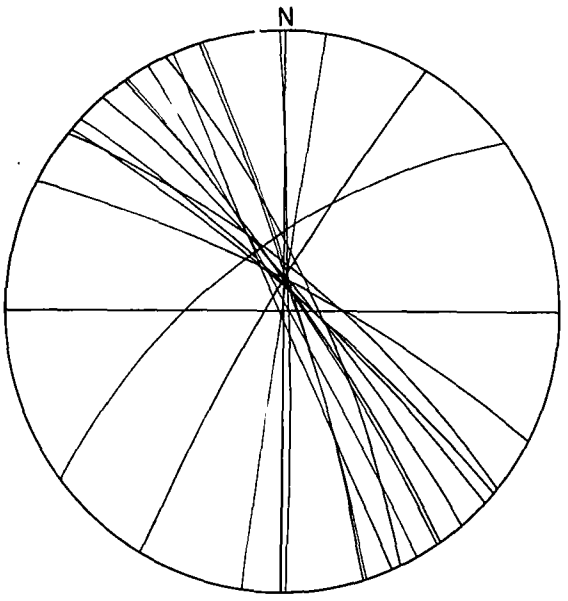


Fig. 7. Circles of remagnetization fitted to the stepwise AF demagnetization data (corrected for the dip of the igneous layering). Remagnetization circles are shown only on the lower hemisphere of the stereographic projection.

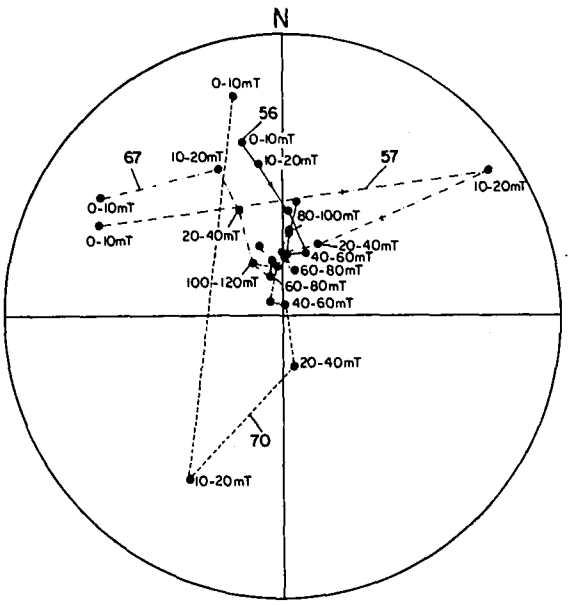


Fig. 8. Difference vectors, removed between successive AF demagnetization steps, are shown for sites 56, 57, 67 and 70.

the difference vectors converge rapidly to a direction which agrees with the mean magnetization direction given by group B. At still higher AF values, the difference vectors then move very rapidly away from this position, each in its own direction, along circles of remagnetization. This pattern of change in direction by the difference vectors at the intersection point, is similar to examples presented by Hoffman and Day [19]. According to them, this pattern indicates the existence of three magnetization components in the specimens with marginal overlapping of the respective coercivity spectra. Furthermore, at the position where the difference vectors converge and sharply change direction, only one magnetization component has been removed. These positions in Fig. 8 were grouped together and yield the following group statistics: $N = 4$; $D = 357.1°$; $I = 69°$; $\alpha_{95} = 5.8°$; $k = 248$ and again this mean direction differs very little from that of group B.

From these analyses it is clear that the NRM directions of sites are in effect resultant magnetization directions, but that the resultant directions differ very little from the primary magnetization direction, because all the secondary components are small compared to the primary magnetization component. Furthermore, all directions obtained for the primary magnetization component with the

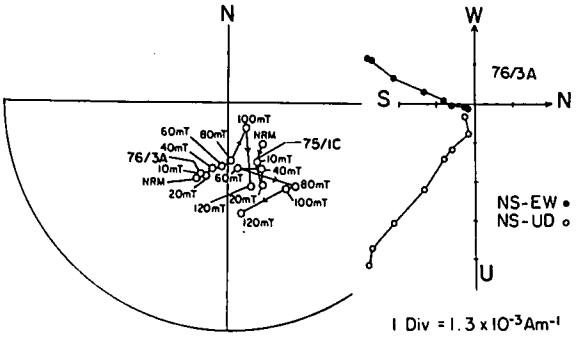


Fig. 9. Directional response of magnetization vectors from specimens 76/3A and 75/1C to stepwise AF demagnetization.

previous methods, fall within the circle of 95% confidence of group B; consequently, the mean direction of this group is acceptable as that of the primary magnetization vector.

A characteristic of all the specimens from sites in group B, is the 'hardness" of the magnetization, manifested by the fact that after demagnetization in an AF of 100 mT, 10–20% of the magnetization remains. This is similar to the behaviour of specimens from certain parts of the main zone in the eastern Bushveld Complex [11].

Specimens from site 75 and 76, which are normally magnetized, have a mean NRM intensity of 4.5×10^{-3} A m^{-1}. This made the results from stepwise AF demagnetization less reliable, but in Fig. 9 the directional response of two specimens from these sites, displays a tendency to converge during AF demagnetization. The approximate convergence zone in Fig. 9 is roughly antipodal to the mean magnetization direction of group B. There is little doubt that the two magnetization directions from sites 75 and 76, obtained by bulk AF demagnetization, do represent a stable and consistent magnetization component.

6. Thermal demagnetization

The directional response of the NRM vector of specimens from group A to continuous thermal demagnetization (Fig. 10) is similar to the response to AF demagnetization. At low temperatures the magnetization directions remain virtually constant, but at higher temperatures the magnetization directions start to change, with the change becoming larger at high temperatures. The vectors do not describe remagnetization circles as in some instances during AF demagnetization but, as during AF demagnetization, the directions diverge more and more as the demagnetization process proceeds. From this it appears as if there are mainly only two magnetization components. The dominant primary magnetization is removed at slightly lower temperatures than the smaller secondary component, which is possibly different for each site.

A typical normalized thermal intensity response curve of a specimen from group A, is shown in Fig. 10B. On this curve two separate Curie points are evident. The lower of these, at approximately 575°C, indicates magnetite as the source of remanent magnetization, while the higher Curie point in the region of 600°C, possibly indicates haematite [20] as the source of magnetization.

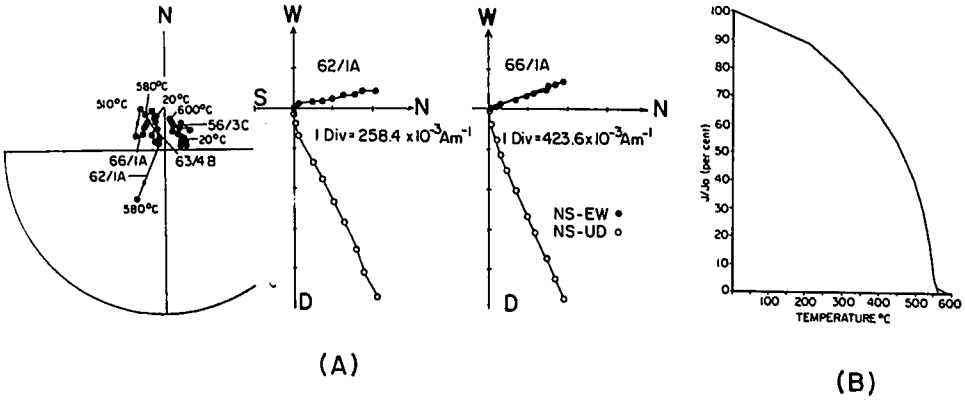


Fig. 10. A. Directional response of magnetization vectors from specimens 56/3C, 66/1A, 63/4B and 62/1A, to continuous thermal demagnetization. B. Normalized intensity response curve of specimen 56/3D during continuous thermal demagnetization.

Thermal demagnetization of specimens from sites 75 and 76 (negative inclinations), could not be carried out due to the low intensity of the magnetization of these specimens.

7. Mineralogy of opaque minerals

All the specimens from sites with reversed magnetization directions (positive inclinations, group B) contain similar opaque minerals. Magnetite occurs as lamellae in both the pyroxene and plagioclase crystals similar to the magnetite in specimens from the main zone in the eastern Bushveld Complex [11]. Magnetite lamellae seem to be less abundant in specimens from the western Bushveld Complex. It must be concluded that the lower Curie point of 575°C, observed for these specimens, can be attributed to these magnetite lamellae. Discrete magnetite grains, although they do occur in the specimens, are very rare.

Ilmenite generally occurs as discrete grains with diameters up to 0.1 mm. These grains commonly exhibit haematite exsolution lamellae. No other haematite is present in the specimens and the second observed Curie temperature of approximately 600°C, could be the result of these exsolution lamellae in the ilmenite.

Sulphide grains occur commonly, but are less abundant than the ilmenite grains. Some of the sulphide grains were identified as pyrrhotite.

The opaque mineralogy of sites 75 and 76, differs considerably from that described for sites in group B. No magnetite grains could be found in either pyroxene or plagioclase crystals and it is noteworthy that the intensity of the NRM of these specimens is two orders of magnitude less than the intensity of the NRM of specimens from sites in group B. Small discrete magnetite grains do occur in these specimens, but are very rare.

A similarity of the opaque minerals in these specimens with respect to specimens from sites in group B, is the occurrence of ilmenite grains with haematite exsolution lamellae. Although no Curie point could be obtained for these specimens due to the low intensity of magnetization, it is possible that the haematite lamellae are the carriers of a primary magnetization in the specimens from sites 75 and 76, as well as the carriers of the hard secondary magnetization in specimens from group B. This will explain the observed magnetic polarity difference which can be the result of a geomagnetic field reversal or a self-reversal of magnetization resulting from a mechanism operating in the haemo-ilmenite grains. However, this possibility can not readily explain the apparent difference of the secondary magnetization components of sites in group B.

8. Summary of results

Fold tests applied to palaeomagnetic directions from the main zone in the western Bushveld Complex, yield results indicating that the main zone possibly acquired its present dip after cooling to temperatures below the Curie points of the respective ferromagnetic components.

The NRM of the main zone comprises a number of magnetization components, but the primary magnetization dominates over the small secondary components to such an extent, that when corrections for these secondary components are applied the mean magnetization direction does not change substantially. This mean magnetization (corrected for the dip of the igneous layering) of the main zone in the western Bushveld Complex is as fol-

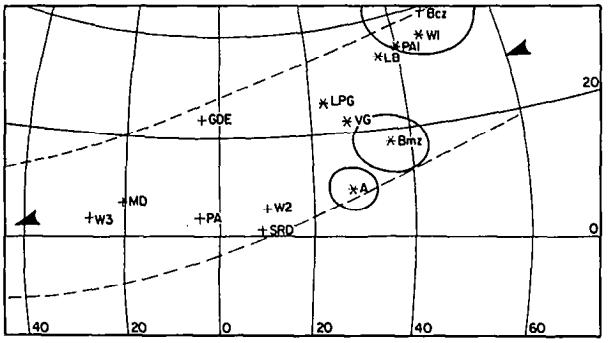


Fig. 11. APW path for Africa with selected pole positions shown for Southern Africa ca. 2070–1700 m.y. *PA1* (2047±11 m.y.) = Phalaborwa pyroxenite [21]; *W1–W3* = Waterberg sediments (stratigraphically arranged poles E, D and C [24]; *VG* (2006±47 m.y.) = Vredefort Granophyre [21,25]; *LPG* (1980±39 m.y.) = Limpopo gneisses [21]; *LB* (1841±276 m.y.) = Losberg Gabbro [21,25]; *GDE* (1895±73 m.y.) = Great Dyke extensions into Limpopo belt [21,26]; *SRD* (1876±68 m.y.) = Sand River dykes [21]; *PA2* (2012±19 m.y.) = Phalaborwa syenite and dolerites [21,22]; *MD* (1811±20 m.y.) = Mashonaland dolerites [22]. *BCZ* (2054±11 m.y.) = critical zone Bushveld Complex [7,23]; *BMZ* (2049±152 m.y.) = main zone eastern Bushveld Complex [7,11]; *B* = main zone western Bushveld Complex this study. * = N poles; + = S poles.

lows—Group B: $N = 32$; $D = 357.8°$; $I = 70.9°$; $\alpha_{95} = 2.8°$; $k = 80$ with corresponding palaeomagnetic pole position: latitude, 9.2°N; longitude, 27.3°E and polar error (dp, dm), 4.2° and 4.9° respectively.

Magnetization directions from two sites in the lower part of the main zone have a different polarity and are approximately antipodal to the mean direction of group B. Although these directions have not been incorporated into group B, one must conclude that the mean magnetization of the main zone in the western Bushveld has a mixed polarity.

The apparent polar wander (APW) path for Africa for the interval circa 2070–1800 m.y. with selected pole positions is shown in Fig. 11. Morgan [21] regards this as a particularly well defined APW segment ~ 80° long with ±10° lateral uncertainty. A comparison of radiometric ages and pole positions (Fig. 11) certainly confirms this with the exception of two poles, namely the Losberg Gabbro (LB) and the younger Phalaborwa (PA2) poles.

The radiometric age of the Losberg Gabbro is 1841 ± 276 m.y. [21]. The relative large uncertainty associated with this age however renders this pole position compatible with other poles on the APW path. The Phalaborwa (PA2) pole represents dolerite dykes and syenite which are younger than the carbonatite and phoscorite of the Phalaborwa Complex. A previous minimum age of 1520 m.y. for the dolerite dykes was given by Morgan and Briden [22], but in a later publication Morgan [21] gives an age of 2012 ± 19 m.y. for the PA2 pole. This age is difficult to accommodate with the PA2 pole position, and the progressive younger radiometric ages manifested from east to west by all other poles on the APW path.

The pole position of the main zone, western Bushveld Complex (pole B, Fig. 11) lies on the APW path to the west of poles BMZ and BCZ which represents the main zone eastern Bushveld Complex [11] and the critical zone of the Bushveld Complex [23] respectively. This implies that emplacement and subsequent cooling of the critical zone was the first phase in the genesis of the layered sequence of the Bushveld Complex. Pole positions (Fig. 11) further indicate that this event took place before or simultaneously with the onset of deposition of Waterberg sediments and the emplacement of the Phalaborwa Complex. This was followed by the emplacement and cooling of the main zone in the eastern Bushveld Complex and only then, the emplacement of the main zone of the western Bushveld Complex.

This is in concordance with the view held by Sharpe et al. [7], based on work in the eastern Bushveld, that the layered rocks were derived from two magma series. The position of pole B (western Bushveld Complex, Fig. 11), however, indicate a possible third influx of magma to form the main zone of the western Bushveld Complex.

Acknowledgements

Numerous companies, institutions and persons made this study possible by providing material aid and granting permission to visit a number of quarries. The author would like to express his gratitude to all those involved, but in particular to the following persons or institutions: The University of Pretoria for a research grant which made this study possible and Prof. G. von Gruenewaldt and Dr. J.S.V. van Zijl for their guidance and helpful suggestions.

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